

CSE4261: Neural Network and Deep Learning

Assignment-4

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Answer: `tf.GradientTape()` vs Tensorflow's `model.fit()`

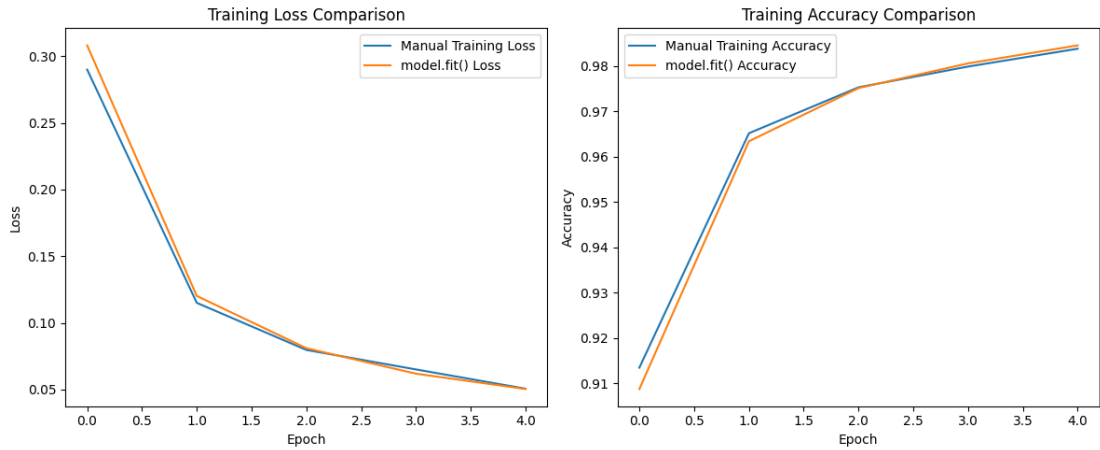
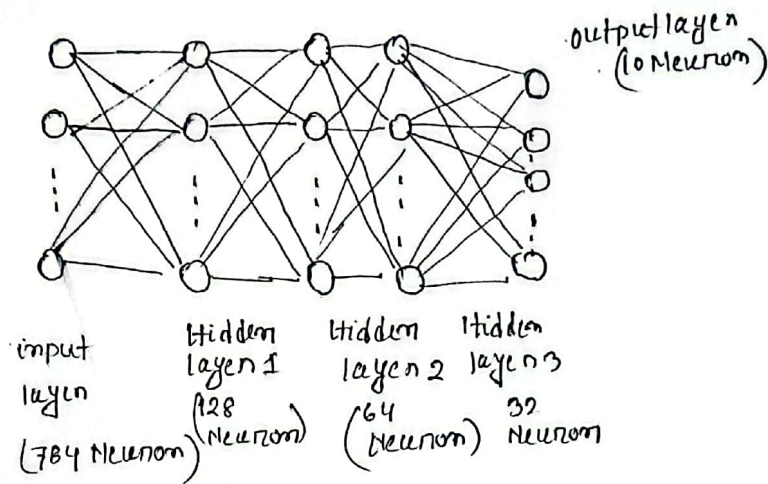


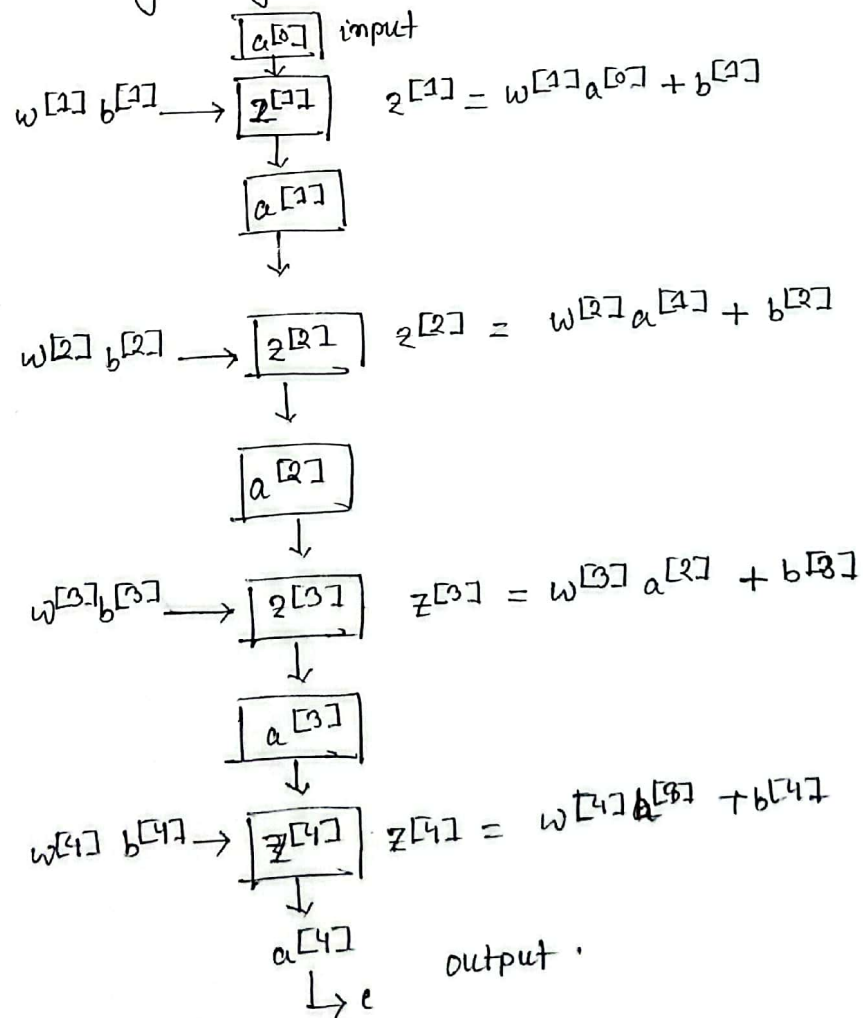
Figure 1: Comparison between `tf.GradientTape()` and Tensorflow's `model.fit()`

When we compare `tf.GradientTape()` with `model.fit()` in TensorFlow, we notice a clear trade-off between control and simplicity. By using `tf.GradientTape()`, we gain full control over each step of the training process, including forward pass, loss calculation, and gradient updates, which is helpful for experimentation and learning. In contrast, when we use `model.fit()`, we benefit from a more concise and user-friendly training interface that handles everything internally. While both approaches allow us to achieve similar performance, we found that `model.fit()` trains faster due to TensorFlow's internal optimizations. However, we prefer `tf.GradientTape()` when we need to understand or customize the training logic deeply.

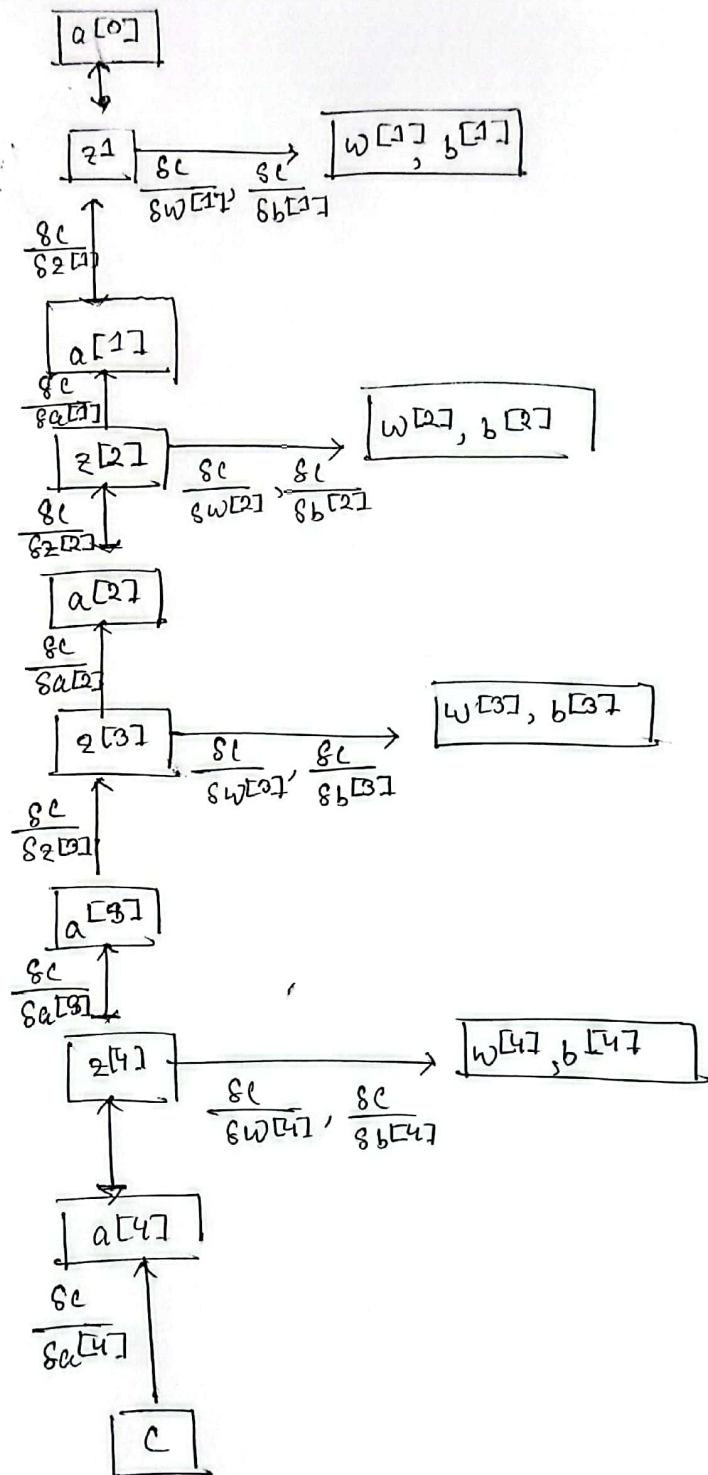
ii Neural Network for classifying 10 class with 3 hidden layers:



ii Forward propagation graph:



Backward propagation graph:



4 Forward pass Equations for layers:

input layer: $a_k^{(0)} = x_k$ for $k = 1, \dots, 784$.

Hidden layer 1:

$$z_j^{(1)} = \sum_{k=1}^{784} w_{jk}^{(1)} a_k^{(0)} + b_j^{(1)}$$

$$a_j^{(1)} = g(z_j^{(1)}) \quad (\text{ReLU})$$

Hidden layer 2:

$$z_j^{(2)} = \sum_{k=1}^{128} w_{jk}^{(2)} a_k^{(1)} + b_j^{(2)}$$

$$a_j^{(2)} = g(z_j^{(2)})$$

Hidden layer 3:

$$z_j^{(3)} = \sum_{k=1}^{64} w_{jk}^{(3)} a_k^{(2)} + b_j^{(3)}$$

$$a_j^{(3)} = g(z_j^{(3)})$$

output layer:

$$z_j^{(4)} = \sum_{k=1}^{32} w_{jk}^{(4)} a_k^{(3)} + b_j^{(4)}$$

$$a_j^{(4)} = g(z_j^{(4)}) \quad (\text{softmax})$$

Loss function:
$$L = - \sum_{j=1}^{10} y_j \log(a_j^{(4)}) \quad (\text{cross entropy})$$

Backpropagation Equation:

output layer:

Error term: $\delta_j^{(4)} = a_j^{(4)} - t_j$

Gradients: $\frac{\delta L}{\delta w_{jk}^{(4)}} = \delta_j^{(4)} a_k^{(3)}, \frac{\delta L}{\delta b_j^{(4)}} = \delta_j^{(4)}$

update: $w_{jk}^{(4)} = w_{jk}^{(4)} - \eta \delta_j^{(4)} a_k^{(3)}, b_j^{(4)} = b_j^{(4)} - \eta \delta_j^{(4)}$

Hidden layer-3:

Error term: $\delta_k^{(3)} = \left(\sum_{j=1}^{10} w_{jk}^{(4)} \delta_j^{(4)} \right) \cdot g'(z_k^{(3)})$

Gradients: $\frac{\delta L}{\delta w_{kl}^{(3)}} = \delta_k^{(3)} a_l^{(2)}, \frac{\delta L}{\delta b_k^{(3)}} = \delta_k^{(3)}$

update: $w_{kl}^{(3)} = w_{kl}^{(3)} - \eta \delta_k^{(3)} a_l^{(2)}, b_k^{(3)} = b_k^{(3)} - \eta \delta_k^{(3)}$

Hidden layer-2:

Error term: $\delta_u^{(2)} = \left(\sum_{k=1}^{32} w_{ku}^{(3)} \delta_k^{(3)} \right) \cdot g'(z_u^{(2)})$

Gradients: $\frac{\delta L}{\delta w_{lm}^{(2)}} = \delta_u^{(2)} a_m^{(1)}, \frac{\delta L}{\delta b_u^{(2)}} = \delta_u^{(2)}$

update: $w_{lm}^{(2)} = w_{lm}^{(2)} - \eta \delta_u^{(2)} a_m^{(1)}, b_u^{(2)} = b_u^{(2)} - \eta \delta_u^{(2)}$

Hidden layer-1:

Error term: $\delta_m^{(1)} = \left(\sum_{u=1}^{64} w_{um}^{(2)} \delta_u^{(2)} \right) \cdot g'(z_m^{(1)})$

Gradients: $\frac{\delta L}{\delta w_{mn}^{(1)}} = \delta_m^{(1)} a_n^{(0)}, \frac{\delta L}{\delta b_m^{(1)}} = \delta_m^{(1)}$

update: $w_{mn}^{(1)} = w_{mn}^{(1)} - \eta \delta_m^{(1)} a_n^{(0)}, b_m^{(1)} = b_m^{(1)} - \eta \delta_m^{(1)}$

ii Backpropagation Equation:

output layer:

$$\text{Error term: } \delta_j^{(4)} = a_j^{(4)} - t_j$$

$$\text{Gradients: } \frac{\partial L}{\partial w_{jk}^{(4)}} = \delta_j^{(4)} a_k^{(3)}, \quad \frac{\partial L}{\partial b_j^{(4)}} = \delta_j^{(4)}$$

$$\text{update: } w_{jk}^{(4)} = w_{jk}^{(4)} - \eta \delta_j^{(4)} a_k^{(3)}, \quad b_j^{(4)} = b_j^{(4)} - \eta \delta_j^{(4)}$$

Hidden layer-3:

$$\text{Error term: } \delta_k^{(3)} = \left(\sum_{j=1}^{10} w_{jk}^{(4)} \delta_j^{(4)} \right) \cdot g'(z_k^{(3)})$$

$$\text{Gradients: } \frac{\partial L}{\partial w_{kl}^{(3)}} = \delta_k^{(3)} a_l^{(2)}, \quad \frac{\partial L}{\partial b_k^{(3)}} = \delta_k^{(3)}$$

$$\text{update: } w_{kl}^{(3)} = w_{kl}^{(3)} - \eta \delta_k^{(3)} a_l^{(2)}, \quad b_k^{(3)} = b_k^{(3)} - \eta \delta_k^{(3)}$$

Hidden layer-2:

$$\text{Error term: } \delta_u^{(2)} = \left(\sum_{k=1}^{32} w_{ku}^{(3)} \delta_k^{(3)} \right) \cdot g'(z_u^{(2)})$$

$$\text{Gradients: } \frac{\partial L}{\partial w_{lm}^{(2)}} = \delta_u^{(2)} a_m^{(1)}, \quad \frac{\partial L}{\partial b_u^{(2)}} = \delta_u^{(2)}$$

$$\text{update: } w_{lm}^{(2)} = w_{lm}^{(2)} - \eta \delta_u^{(2)} a_m^{(1)}, \quad b_u^{(2)} = b_u^{(2)} - \eta \delta_u^{(2)}$$

Hidden layer-1:

$$\text{Error term: } \delta_m^{(1)} = \left(\sum_{u=1}^{64} w_{mu}^{(2)} \delta_u^{(2)} \right) \cdot g'(z_m^{(1)})$$

$$\text{Gradients: } \frac{\partial L}{\partial w_{mn}^{(1)}} = \delta_m^{(1)} a_n^{(0)}, \quad \frac{\partial L}{\partial b_m^{(1)}} = \delta_m^{(1)}$$

$$\text{update: } w_{mn}^{(1)} = w_{mn}^{(1)} - \eta \delta_m^{(1)} a_n^{(0)}, \quad b_m^{(1)} = b_m^{(1)} - \eta \delta_m^{(1)}$$